

Frontier AI Companies Are Racing to Automate AI Research

Severin Field, Independent Researcher and Visiting Fellow at IAPS. Views are my own.

Key Takeaways

- This past year, I interviewed 25 research scientists at OpenAI, Anthropic, Google DeepMind, Meta, Princeton, UC Berkeley, and Stanford about AI systems being used to automate the process of AI research itself.^{1,2} I call this “recursive self-improvement” (RSI): an AI system good enough at AI development that it can design the next, more powerful version of itself, and so on.
- Making AI better at programming and AI research and then using it to build their successor has become an official plan at Anthropic and OpenAI.^{3,4,5} They say this plainly; interviewing them confirms this vision.
- 20 of 25 interviewees identified automating AI R&D among the most severe and urgent risks: AI capabilities could improve much faster than our ability to oversee, steer, or govern them. Fully automating AI R&D could set off RSI.
- The US government needs more AI capacity and visibility into AI development as we approach critical milestones.

The Most Capable Models May Never Be Public

- Over half of the interviewees expected the most powerful models to increasingly be kept behind closed doors. The recent OpenAI/HuggingFace incident came from an internal-only model the public was not aware of.

The Path to Full Automation

- Recursive self-improvement is increasingly a regular company-wide discussion topic at frontier AI companies, including lead researchers and CEOs.
- Amodei: making models “good at coding and good at AI research,” using them “to produce the next generation of models... to create a loop.”⁵ Altman’s stated plan: an automated AI researcher that increasingly automates research itself.⁴
- Little consensus among interviewees on timelines or mitigations, but they generally expected a progression from coding assistants that need oversight, to systems that meaningfully contribute to research under human direction, to systems that run research and development cycles independently.
- The “Task Horizon Benchmark” maintained by METR,⁶ an independent AI evaluation organization, was consistently named the best public measure of progress. AI systems can now autonomously handle software engineering tasks that take a skilled human around 4 hours — and the time horizon doubles roughly every 7 months. If this rate holds: ~1 month of skilled software engineer labor by 2030, ~1 year a year later.
- Half expected they already have a clear, continuous trajectory towards full automation — achievable almost entirely through scaling investment in AI hardware and algorithmic progress. Even skeptical interviewees — who believe a fundamental breakthrough is needed, such as “creativity” — agreed AI companies are racing for the “missing ingredients.”

Those Closest to RSI Are Asking for Guardrails

- In July 2026, 1,224 employees of frontier AI companies — including the chief scientists of OpenAI, Meta, and Thinking Machines, and Anthropic’s CEO and co-founders — signed an open statement⁷ warning their companies “could be close to automating AI research” and requesting the U.S. government “support an international effort to develop the ... tools needed to deliberately pace the frontier of automated AI development.”
- OpenAI and Anthropic endorsed the statement; both have said publicly the world may need to slow down as capabilities approach critical thresholds.

Recommendations From Interviewees

- Interviewees were split on thresholds that trigger government response. One problem they flagged: the more precisely you define a threshold, the easier it is to enforce, but the worse it captures a dynamic risk that is inherently uncertain. Interviewees universally agreed on two priorities: increasing visibility and capacity.
- **Visibility:** how much the U.S. government and general public know about AI developments, especially given these might be kept internal. Often suggested: mandatory reporting requirements. Might also include:
 - Ensure government access to internal-only deployments: The America First Policy Institute warns undisclosed models available only to company staff create “a loophole in government testing and access.”⁸
 - Hold public hearings: this could include CEOs and senior researchers who see frontier AI systems months before the public. They could be asked about internal deployments and forecasts.
 - Tracking AI progress: public benchmarks or standing interview programs. Congress and the public are currently reliant on AI companies giving independent evaluators (e.g. METR) advance model access.
- **Capacity:** the government’s ability to forecast AI capabilities and build safeguards amid rapid AI progress.
 - One particularly useful capacity: the science surrounding AI treaty verification. This is because the U.S. cannot credibly propose, enter, or enforce any international agreement (e.g. nonproliferation with the PRC) without it.^{9,10}

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Notes

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